



Siamese neural network architecture for homoglyph attacks detection

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Abstract

Primarily an adversary uses homoglyph or spoofing attack approach to obfuscate domain name, file name or process names. This approach facilitates to create domain name, file name or process names which look visually homogeneous to legitimate domain name, file name or process names. This paper introduces Siamese neural network architecture which uses the application of recurrent structures with Keras character level embedding to learn the optimal features by considering an input in the form of raw strings. For comparative study, various recurrent structures are used. The performances obtained by recurrent structures are almost closer. However, the proposed method performed well in comparison to the existing methods such as Edit Distance, Visual Edit Distance and Siamese convolutional neural networks.

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Keywords: Homoglyph; Spoofing; Deep learning; Siamese neural networks; Recurrent structures

1. Introduction

Cyber-attackers have used new types of attack methodology to infiltrate information and communication technologies (ICTs) systems. The number of attacks and complexity of attacks continued to grow exponentially for ICT systems in recent days. One of the well-known attacks is homoglyph or spoofing to obfuscate domain name, file name or process names with the motivation for malicious activities [1]. Mostly, cyber attackers use simple character substitution method for example 1 for *l*, 0 for *O* etc. The character substitution can also include Unicode characters that look homogeneous to most common ASCII characters. Another method is to append characters at the end of a string that looks similar to valid strings. Overall, the primary aim is to make the domain or process name remain undetected by security organizations and network admin.

One of the most commonly used methods for identifying spoof attack is to estimate the Edit Distance using Damerau–Levenshtein [2]. It means the similarity between each new domain or process name and a set of each domain or process name is estimated using the Edit Distance. Generally, it gives

the measure based on number of edits such as insertions, deletions, substitutions or transpositions. On the basis of threshold for distance measure, a string is flagged as possible spoof. The performance obtained by this method is very less in real-time environment. Another method is based on custom Edit Distance which gives importance to visual similarity of substitutions. This gives a smaller Edit Distance for a visually similar character and performed better than the standard Edit Distance functions [3,4]. However, the performances obtained by this method are relatively moderate compared to standard Edit Distance functions. Additionally, both Visual Edit Distance and Edit Distance based methods are based on manual feature engineering. This is considered as one of the daunting tasks because it required extensive domain knowledge. Moreover, these are more vulnerable in an adversarial environment.

The methods based on deep learning outperformed the existing methods in several applications related to natural language processing (NLP), speech recognition, and computer vision applications [5]. In this paper, the application of deep learning transformed into homoglyph detection of domain and process name spoofing [6]. This method performed better than the existing methods based on Edit Distance and Visual Edit Distance. To extract similarity, the application of siamese convolutional neural networks (Siamese-CNNs) is used. The names of domain and process are converted into

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image by taking the ASCII value for character. These values are passed into Siamese-CNNs for obtaining feature vectors that best distinguish between similar and dissimilar domain and process names. In this paper, this work leverages the application of Keras character level embedding to convert the characters into numeric form and used Siamese recurrent structures network (SRSN) and compared with the SCNNs and other classical based methods based on Edit Distance and Visual Edit Distance. The rest of the paper is organized as follows: Section 2 provides a short review on related works of spoofing detection. Section 3 includes the proposed architecture. Section 4 discusses experimental analysis, observations and results. Section 5 includes conclusion.

2. Related works

There are various methods that exist for effective string matching. Few works were given importance to the computational time for string matching [7,8]. In [9] is proposed string matching algorithm that accounts for enhancing the nature of nearest neighbor searches. To enhance the system based on Edit Distance, the features related to visual similarity of characters are used. As a result, visual similarity enhanced the spoofing detection rate in comparison to the normal Edit Distance based methods. However, these Edit Distance based methods [3,4] are not completely applicable for domain or process name spoofing in real-time environment. In [10] authors discussed a detailed study on the characteristics of visually appear strings. In [11] authors discussed the application of n-gram text representation with machine learning algorithms such as SVM and random forest for typosquatting detection at domain level. To avoid manual feature engineering, [6] proposed Siamese-CNNs for domain or process name similarity detection. The strings are converted into image representation by transforming each character into an ASCII value representation.

3. Proposed architecture

The proposed architecture for homoglyph or spoofing attacks detection of domain and process names is shown in Fig. 1. It uses the application of Siamese neural network architecture [12] to extract the similarity that exists between a set of domain names or process names with the aim to detect homoglyph or spoofing attacks. During training, the architecture takes a set of domain or process names along with a similarity score to the proposed architecture. This can be mathematically represented as $\{(DP_i, DP'_i, S_i)_{i=1}^n\}$, where DP be a domain or process name, DP' be a spoof of domain or process name, $S \in \{0, 1\}$, 0 for similar and 1 for dissimilar. The proposed architecture contains two identical neural networks. Each neural network takes distinct input by taking into account the shared weights and outputs of neural network that are merged by a simple comparative energy function. Euclidean distance is used as comparative energy function and defined mathematically as $D_w(DP_1, DP_2) = E(N_w(DP_1), N_w(DP_2)) = \|N_w(DP_1) - N_w(DP_2)\|_2$, where (DP_1, DP_2) a pair of strings,

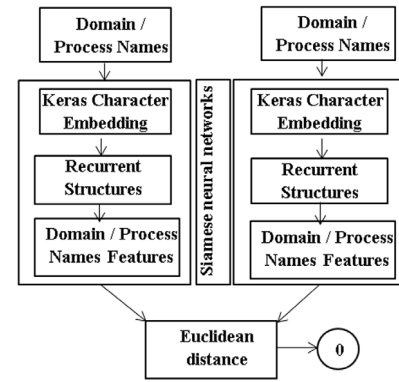


Fig. 1. Proposed architecture: An overview of training Siamese Neural network. The network accepts spoofing pairs of domain names or process name and the recurrent structures are training such that Euclidean distance between their respective features is 0.0.

N_w denotes functions parametrized by w . The proposed architecture aims to learn w such that $D_w(DP_1, DP_2)$ is 0 for similar strings and 1 for dissimilar strings. To minimize w , small distance for similar inputs and large distance for dissimilar inputs is considered by [8]. In this work, to achieve this, contrastive loss function defined by [11] is used. Contrastive loss is defined mathematically $(i - S) + \frac{1}{2}(D_w)^2 + (S)\frac{1}{2}\{\max(0, m - D_w)\}^2$, where D_w defines the Euclidean distance between $D_w(DP_1)$ and $D_w(DP_2)$, $S \in \{0, 1\}$ 0 for similar and 1 for dissimilar, $\max$ is maximum function and m is a margin value. The margin value is mostly greater than 0 so that the similar sets to 0 and dissimilar sets to be at least m .

Each identical neural network of proposed architecture contains an embedding layer followed by Siamese recurrent structures network (SRSN) such as recurrent neural network (RNN), identity recurrent neural network (IRNN), long short-term memory (LSTM) and Gated recurrent unit (GRU). Keras character level embedding takes 3 parameters. They are dictionary size to 200, maximum length of domain name to 100 and embedding vector length to 128 in the case of domain name and dictionary size to 250, maximum length of domain name to 85 and embedding vector length to 128 in the case of process name. In recurrent structures the number of units is set to 128 and *RMS Prop* optimizer on batches of 16 string samples.

4. Experimental analysis, observations and results

All deep learning architectures are implemented using TensorFlow,¹ Scikit-learn² with Keras³ library. The experiments are run on GPU enabled computers. To evaluate the performance of Siamese recurrent structures network (SRSN), the subset of domain and process name data set is used [6]. The legitimate samples of process name contain Edit distance (Damerau–Levenshtein) less than or equal to 3. This distance is very small and helps to differentiate the visual similarity

¹ <https://www.tensorflow.org/>.

² <https://scikit-learn.org/stable/>.

³ <https://keras.io/>.

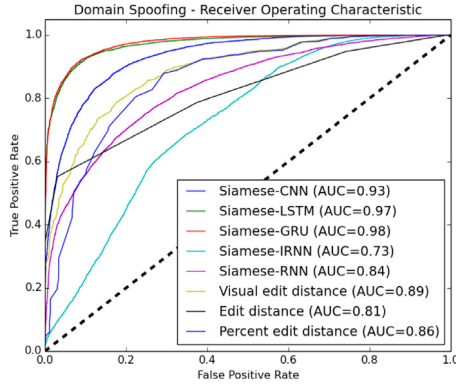


Fig. 2. ROC curve for domain name spoofing.

from Edit distance similarity of string samples. This type of data set facilitates to identify simple algorithms based on distance functions. The process name is composed of at least 4 characters. These names are file name and as well as .dll names chosen from the National Software Reference Library (NSRL) [13]. The spoofing pairs are generated by following character substitution which includes both ASCII and Unicode. Same method is followed for domain name except including constraint on the distance value. This is mainly due to the reason that there may be possibility that the legitimate domain name samples may contain distance value less than or equal to 4. Both the data set contains 20,000 for training, 15,000 for validation and 15,000 for testing. Each sample of the data set contains a pair of domain names or process names with a value 0 (similar) or 1 (dissimilar).

All recurrent structures are run till 10 epochs. To control the training accuracy, validation data set is used. This helped to avoid over fitting. After training, the trained models of SRSN are evaluated on test data set for both the domain and process name spoofing. The performances obtained by various models are evaluated in terms of area under the Receiver operating characteristic (ROC) area under curve (AUC) using the Receiver operating characteristic curve and reported in Table 1. The ROC curve for domain name and process name is shown in Figs. 2 and 3 respectively. ROC curve is drawn using the false positive rate on X axis and true positive rate on Y axis with threshold 0.5. TPR is the ratio between the true positive (TP) values and the total true positive and false negative (FN) values. FPR is the ratio between the false positive (FP) values and the total false positive and true negative (TN) values. The TP, TN, FP and FN defined as follows: TP defines the number of similar string samples either domain or process name correctly classified as similar, TN defines the number of dissimilar string samples either domain or process name correctly classified as dissimilar, FP defines the number of dissimilar string samples either domain or process name incorrectly classified as similar and FN defines the number of similar string samples either domain or process name incorrectly classified as dissimilar. The proposed method based on LSTM and GRU recurrent structures performed better than the existing methods such as Edit Distance, Visual Edit Distance and Siamese convolutional neural networks Siamese-CNNs.

Table 1

Performance of the model — Percent Edit Distance: Edit distance normalized by domain or process name length.

Method	ROC-AUC
Domain name spoofing	
Siamese-CNN [5]	0.93
Siamese-RNN (Proposed method)	0.84
Siamese-IRNN (Proposed method)	0.73
Siamese-LSTM (Proposed method)	0.97
Siamese-GRU (Proposed method)	0.98
Visual Edit Distance [6]	0.89
Edit Distance [6]	0.81
Percent Edit Distance [6]	0.86
Process name spoofing	
Siamese-CNN [5]	0.77
Siamese-RNN (Proposed method)	0.79
Siamese-IRNN (Proposed method)	0.58
Siamese-LSTM (Proposed method)	0.95
Siamese-GRU (Proposed method)	0.95
Visual Edit Distance [6]	0.43
Edit Distance [5]	0.51
Percent Edit Distance [6]	0.44

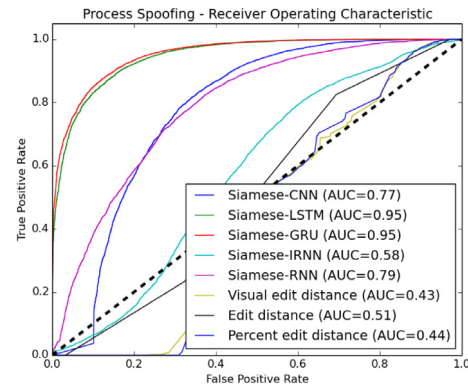


Fig. 3. ROC curve for process name spoofing.

The performance of the recurrent structures such as RNN and IRNN is considerably less compared to all the existing methods such as Edit Distance, Visual Edit Distance and Siamese-CNNs in domain spoofing. In the case of process spoofing experiments, the performance of RNN and IRNN models is better than the existing classical methods such Edit Distance and Visual Edit Distance and less than the Siamese-CNNs. The primary reason may be due to the fact that the both the models do not maintain a memory to keep significant information. However, the performance of both RNN and IRNN can be enhanced by following hyper parameter selection method to select best parameters. Hyper parameters play a significant role in obtaining better performance [5]. Additionally, the model performances are tested on the subset of the data set [6]. Testing the performances on the entire full data set remained as a significant direction towards future works. This can further enhance the performances of various SRSN towards detecting the Domain and Process name spoofing.

Table 2
Detailed parameter details of Siamese models.

Method	Domain name spoofing	Process name spoofing
	Total parameters	Total parameters
Siamese-CNN [6]	148,832	148,832
Siamese-RNN (Proposed)	58,496	58,496
Siamese-IRNN (Proposed)	58,496	58,496
Siamese-LSTM (Proposed)	157,184	157,184
Siamese-GRU (Proposed)	124,288	124,288

The parameter details of all the SRSN are reported in Table 2. The detailed configuration parameter details are available at.⁴ The proposed models are computationally inexpensive compared to the Siamese-CNNs [6]. As reported in Table 1, the performance obtained by Siamese-LSTM and Siamese-GRU is same and in domain name spoofing Siamese-GRU outperformed Siamese-LSTM. However, the Siamese-LSTM based spoofing detection in domain and process names is computationally expensive compared to Siamese-GRU and Siamese-LSTM contains more number of parameters compared to Siamese-GRU.

5. Conclusion and future work

This paper introduces Siamese recurrent structures network (SRSN) for domain or process name homograph or spoofing attack detection. Based on dictionary, characters of domain or process names are converted into index, numeric representation and passed into SRSN. SRSN uses Keras character level embedding following various recurrent structures for feature extraction. SRSN model used Euclidean distance function to estimate the similarity of feature vectors of strings extracted by recurrent structures during training. The Euclidean distance helps to extract optimal features in such a way that the distance is small for feature vectors which contain similar strings and distance is large for dissimilar strings. Results obtained by proposed method performed better than the Edit Distance, Vi-

sual Edit Distance and Siamese convolutional neural networks. However, the same experimental analysis can be tested on the full data set. Additionally, the concept of adversarial approach can be mapped into process or domain name. These remain as a significant direction towards future works.

Declaration of competing interest

The authors declare that there is no conflict of interest in this paper.

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⁴ <https://github.com/vinayakumarr/homoglyph-Siamese>.